Task 3: The Overfitting Astronaut

Step 1: My Rule (the model)

Say YES only if ALL of these are true:

1. The person is wearing a **blue shirt**, **and**
2. The sign is **white** with the word **"FOLLOW" in black marker**, **and**
3. The background is **bright/sunny** (or clear sky or rocky ground), **and**
4. The sign is held **right-side up** (straight or slightly tilted).

Otherwise, say NO.

Step 2: Testing the Rule

Photo	Description	My Rule	Reasons
A	White EVA suit, red "FOLLOW", bright sun	NO	Not a blue shirt, and the marker is red, not black
B	Blue shirt, upside-down "FOLLOW", cloudy	NO	Sign is upside down, and the day is cloudy, not bright
C	Blue shirt, "STOP" in black, bright sun	YES	Blue shirt, black text, and bright sun all match
D	White EVA suit, hand-drawn arrow, bright sun	NO	Not a blue shirt, and there is no word "FOLLOW"

Step 3: Comparison with the Truth

Photos	My answer	truth	result
A	NO	FOLLOW	WRONG
B	NO	FOLLOW	WRONG
C	YES	NOT FOLLOW	WRONG
D	NO	FOLLOW	WRONG

Reflection Answers

1. My rule got **0 out of 4** correct. It failed on every photo. It rejected three real Follow signs (A, B, D) and accepted a STOP sign (C), which is the most dangerous kind of mistake because the astronaut would be told "Follow" when the sign actually says "Stop."

2. The rule focused on things that are not the sign: the **blue shirt**, the **marker colour**, and the **bright sunlight**. In all six training photos the astronaut wore the same blue shirt, so the model treated "blue shirt" as if it meant "Follow." In A and D the astronaut wore a white EVA suit, so the rule said NO even though the sign was valid. This is **data bias**: the training data was not diverse, so the model learned a coincidence instead of the real concept. The same bias is why C (blue shirt, but "STOP") was wrongly accepted.

3. The rule was too specific to the exact conditions seen in training: an upright sign and a bright sunny day. In B the sign was upside down and the sky was cloudy, so the rule rejected it, even though the word "FOLLOW" was still there. This is **overfitting**: the model memorised irrelevant details (orientation, lighting) instead of learning the general idea that the meaning of the sign is what matters.

4.

- **Photo 1:** A **white EVA suit** (or other clothing), so the model learns that clothing doesn't matter.
- **Photo 2: Different lighting and weather** (cloudy, dusk, shadow) so the model learns that lighting don't matter.
- **Photo 3:** A **different sign design** (an arrow, a different marker colour, a different sign colour), plus **negative examples** labelled "NOT Follow" (such as a STOP sign), so the model learns what the sign actually has to contain.

5. If the astronaut wears a **white EVA suit**, the model would likely fail to recognise valid signs, because it learned "blue shirt = Follow." The rover could ignore the signal, or even act wrongly on a sign it should not obey, and lose points.

A better training set would:

- Include **many clothing types**, including the white EVA suit.
- Cover **varied lighting and weather**: bright sun, shadow, cloud, dusk.
- Use **varied backgrounds** (desert, rock, sky) and **camera distances and angles**.
- Include **different sign designs** allowed by the rules, such as words and arrows, in different colours and orientations.
- Include **negative examples** (STOP signs, blank signs, random objects) so the model learns the difference.
- Be **tested on photos it has never seen** before competition day.